

The Conservation Laws of the Standard Model as the Symmetries of One Geometry

Standalone companion to the flagship — DRAFT, Keeper, 2026-07-18. Every conserved charge is a symmetry — continuous or topological — of D_{IV}^5 .

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Abstract

We account for the conservation laws of the Standard Model as symmetries of a single bounded symmetric domain, $D_{IV}^5 = SO(5,2)/[SO(5) \times SO(2)]$. The continuous conservation laws are the Noether currents of the domain's isometry group: energy, momentum, and angular momentum from the spacetime isometries; electric charge from the $SO(2)$ complex structure; and weak isospin, hypercharge, and color from the domain's three division-algebra structures. Their count is $20 = g + c_3$, both invariants of the domain. The discrete conservation laws are *topological* — beyond Noether: baryon number is a trefoil-knot count, lepton number a winding count, and generation number a cycle count. This explains what the Standard Model leaves unexplained — why baryon and lepton number are conserved at all (topology protects them) and how they are violated (only by discrete topology change: the $\Delta L=2$ Majorana neutrino mass). Finally, the *pattern* of which symmetries hold exactly and which are broken is itself a prediction: CPT is exact, while parity, CP, and scale are broken by mechanisms the geometry derives (an odd embedding dimension; the Jarlskog phase; the mass scale).

1. Principle

Noether's theorem turns each continuous symmetry into a conserved current; a domain's discrete symmetries give C, P, T; and its *topology* gives conserved counts no continuous symmetry produces. The isometry algebra of D_{IV}^5 is $so(5,2)$ (dimension $21 = N_c + g$), forced by the requirement of three colors, three generations, and complex dimension five. Conservation is therefore not an input — it is the shadow of the domain's symmetry and its topology.

2. The continuous (Noetherian) conservation laws

Each is a specific generator of $so(5,2)$ or of the isotropy $K = SO(5) \times SO(2)$ (T1945; the 21 generators verified as exact isometries, Elie):

conservation law	#	origin (symmetry of D_{IV}^5)	status
Energy	1	time translation	exact
Momentum	$3 = N_c$	spatial translation	exact
Angular momentum	$3 = N_c$	$SO(3)$ rotation	exact
Electric charge	1	the $SO(2)$ complex structure J	exact
Weak isospin	$3 = N_c$	$SU(2)_L \subset SO(5)$	EWSB-hidden
Weak hypercharge	1	$U(1)_Y$	EWSB-hidden
Color	$8 = C_2 - N_c$	$SU(3)$ (compact-dual Q^5)	exact

Continuous total = 7 (spacetime) + 13 (gauge) = 20 = $g + c_3$, both BST integers (c_3 = the third Chern integer of the compact dual Q^5). The Lorentz subalgebra has dimension $6 = C_2$. [derived tally; the domain fixes which generator gives which law]

3. The topological (non-Noetherian) conservation laws

These are protected by *homotopy*, not by any continuous symmetry — Noether cannot produce them (T1945/T1929):

conservation law	topological origin	why beyond Noether
Baryon number B	trefoil (3-crossing) knot count ($N_c = 3$ forced)	a knot cannot be continuously unknotted
Lepton number L	SO(5) winding count (leptons colorless)	a winding number is a homotopy class
Generation number	Q^5 cycle count	cycle count on the compact dual

4. What Noether cannot do — the topological origin explains the SM's accidents

In the Standard Model, baryon and lepton number are *accidental* global symmetries: nothing forces them, and Noether supplies them only as after-the-fact U(1)s. BST gives them a cause: - Why they are conserved: topology protects them (a knot, a winding). - How they are violated: only by *discrete topology change*. B+L is violated by electroweak sphalerons (which preserve B–L); L (and B–L) is violated by exactly two units by the neutrino Majorana mass — the $\Delta L=2$ event that makes neutrinoless double-beta decay occur. Baryon number, the trefoil count, is *absolutely* conserved: the proton is stable, $\tau_p = \infty$ (there is no grand-unified scale into which the color integer could dissolve). This is the content the Standard Model lacks: not just *that* B and L are conserved, but *why*, and *how* they break.

5. The exact/broken pattern is itself a prediction

BST reproduces not only the conservation laws but which hold exactly and which break, and why: - Exact: energy, momentum, angular momentum, electric charge, color, CPT (from the connected isometry group), and baryon number. - Broken by derived mechanisms: parity (the volume element is central because $g = 7$ is *odd* — a chirality lock, K729); CP (the Jarlskog phase); scale/dilatation (broken by the mass ruler — exact only on the conformal, massless boundary); lepton number (the $\Delta L=2$ Majorana mass). Every breaking has a *named geometric source*. The pattern is a downstream prediction of the geometry, not an input.

6. Falsifiability

- Baryon number = trefoil $\Rightarrow$ the proton is absolutely stable. A confirmed proton-decay event at *any* lifetime (Super-K / Hyper-K) falsifies BST.
- Lepton number violated by $\Delta L=2$ (Majorana) $\Rightarrow$ neutrinoless double-beta decay occurs, $|m_{\beta\beta}| \in [1.44, 3.63]$ meV — a detection supports BST; a confirmed null below ~ 1 meV (with $m_1 = 0$ established) falsifies it. These two — proton stability and a $0\nu\beta\beta$ signal in a specific window — are the sharp topological-conservation predictions.

7. What is not claimed

- The explicit current forms ($j^\mu_\xi = T^{\mu\nu}\xi_\nu$ for isometries; the internal currents) require the explicit Bergman–Dirac Lagrangian; the *inventory* (which symmetry gives which law) is complete, the closed-form currents are framework-pending.
- Color conservation rides the color-structure tier (SU(3) hosted on the compact dual; confinement is derived, the full dynamics is open).
- The continuous tally $20 = g + c_3$ is a bookkeeping identity, not a representation dimension (we do *not* claim the unrelated appearance of $27 = 7+13+\dots$ as structural).

8. Conclusion

The conservation laws of the Standard Model are the symmetries of one domain: its isometries give the continuous charges ($20 = g + c_3$), and its topology gives three charges no continuous symmetry can — with the topological origin explaining what the Standard Model leaves as accidents, and the exact/broken pattern falling out as a prediction. Conservation is the shadow of the geometry.

Draft status: complete as a short standalone; companion to the flagship, the α standalone, and the Five-Absence falsifier. Needs: Cal referee on the $20 = g + c_3$ split and the topological identifications; the explicit currents inlined once the Bergman–Dirac Lagrangian (LAG-1) is available. Dimension counts verified (`conservation_inventory.py`).

— Keeper, 2026-07-18. Conservation = Noether on $so(5,2)$ ($20 = g + c_3$) + three topological charges (baryon = trefoil, lepton = winding, generation = cycle — beyond Noether). Topological origin explains the SM’s B/L accidents; exact/broken pattern is a prediction (CPT exact; P/CP/scale/L broken by derived mechanisms). Falsifiers: proton stability + $0\nu\beta\beta$. See [\[\[Keeper_Conservation_Laws_Inventory_from_the_isometry_group_2026-07-18\]\]](#), [\[\[BST_FLAGSHIP_The_Standard_Model_as_Representation_Theory_of_D_IV5_DRAFT_2026-07-18\]\]](#).